THOMAS MORE

Data Science Fundamentals

Recap

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Before we get started…

Practicalities

Learning objectives

- You understand and apply different types of regression, know when and how to use them, can execute them and put them in to practice as well as interpret the results.
- You recognize overfitting in regression models and learn techniques to prevent overfitting.
- You learn about different clustering techniques and understand when and how to use them for grouping data.
- You gain an understanding of dimensionality reduction techniques and learn how to apply them to compress and visualize data.

Course material

- **Canvas**
  - Slides & exercises from classes
- **Online**
  - https://www.w3schools.com/datascience/
- **Software**
  - JupyterLab (or work via Anaconda/VS Code if you prefer)
- **The Internet**



Course breakdown

- **Course Structure:**
  - Lectures: 3 hours per week (theory and exercises)
  - Attendance: Mostly not compulsory, but strongly recommended.

- **Examination Breakdown:**
  - 30%: Continuous assessment (mandatory exercises)
  - 70%: Final exam (oral, mostly theoretical with some practical elements)

Is it difficult to pass?

It’s perfectly doable **if you put in the work**

- Come to class
- Participate in the lessons
- Make your exercises



How to be an ideal student

- **Attend classes** - even when you party hardy
- **Show respect** - towards your instructor and fellow students
- **Be punctual** - or you'll find the door closed
- **Meet deadlines** - late submissions will earn a solid zero
- **Have a question?** - just ask! I don't bite
- **Know it all?** - help your classmates, they might repay you with pizza
- **Do your own work!** - shortcuts will lead to a short academic career ;)

Syllabus

Week - 1	Introduction & Python/JupyterLab Setup	Week - 9	Easter Holiday
Week - 2	Regression - Linear regression	Week - 10	Easter Holiday
Week - 3	Regression - Polynomial regression	Week - 11	Clustering - Hierarchical clustering
Week - 4	Classification - Logistic regression	Week - 12	Dimensionality reduction: Feature Analysis - PCA
Week - 5		Week - 13	Regularization: Lasso regression
Week - 6		Week - 14	Recap: Theory + Excercises
Week - 7		Week - 15	Exam Preperation + Q&A
Week -			



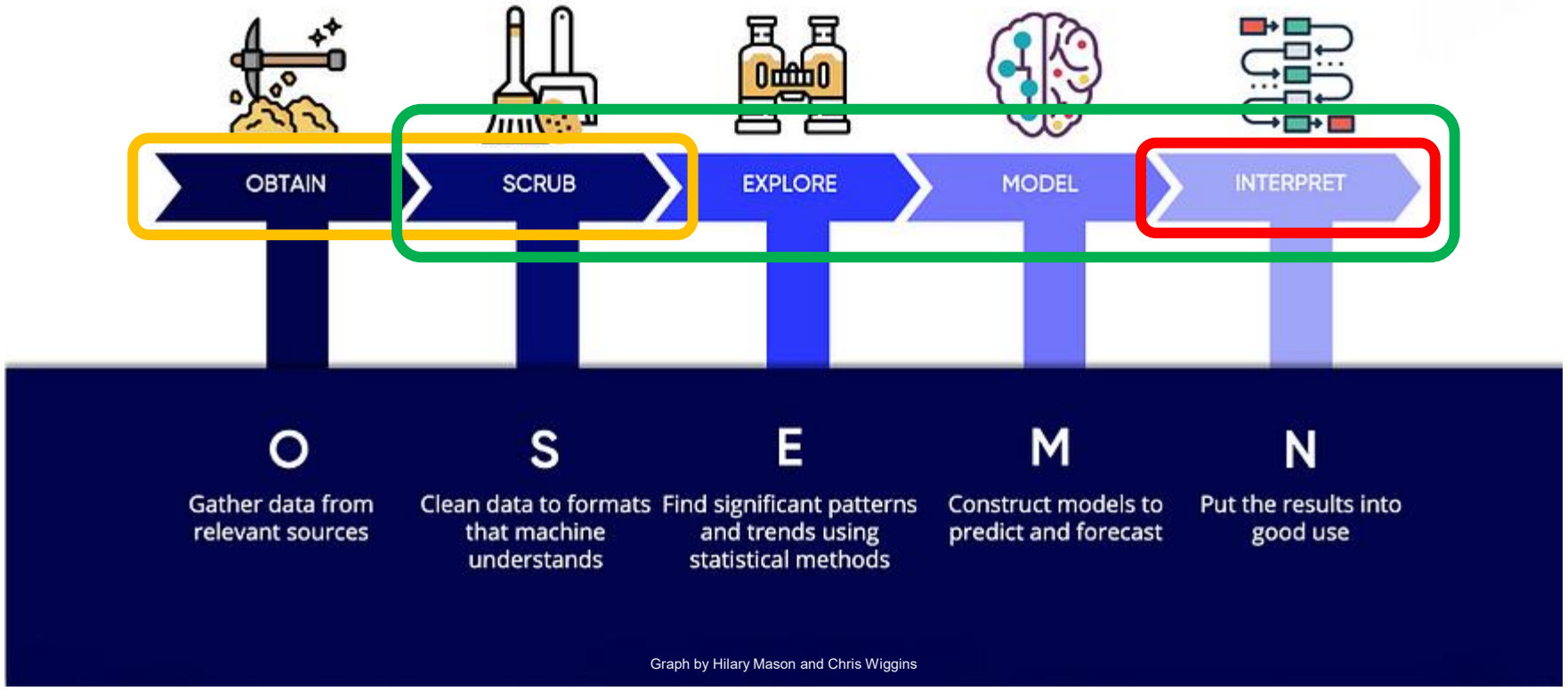
(subject to change)

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Now we get to the fun part

Recap

The data science process



Legend:

Data engineer

Data analist

Data scientist

The data science process



Data cleaning

Data exploration

Dimensionality reduction

Model preparation

Model training

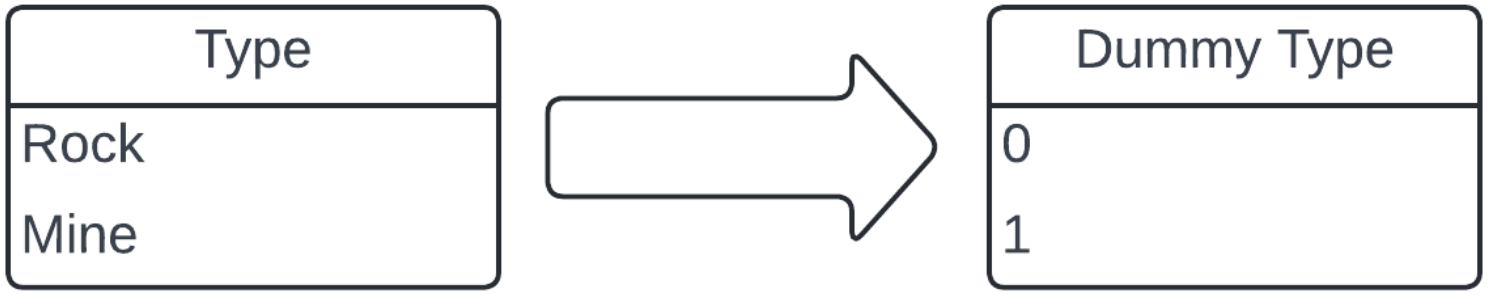
Evaluation metrics



1 – Data cleaning

Dummy variables

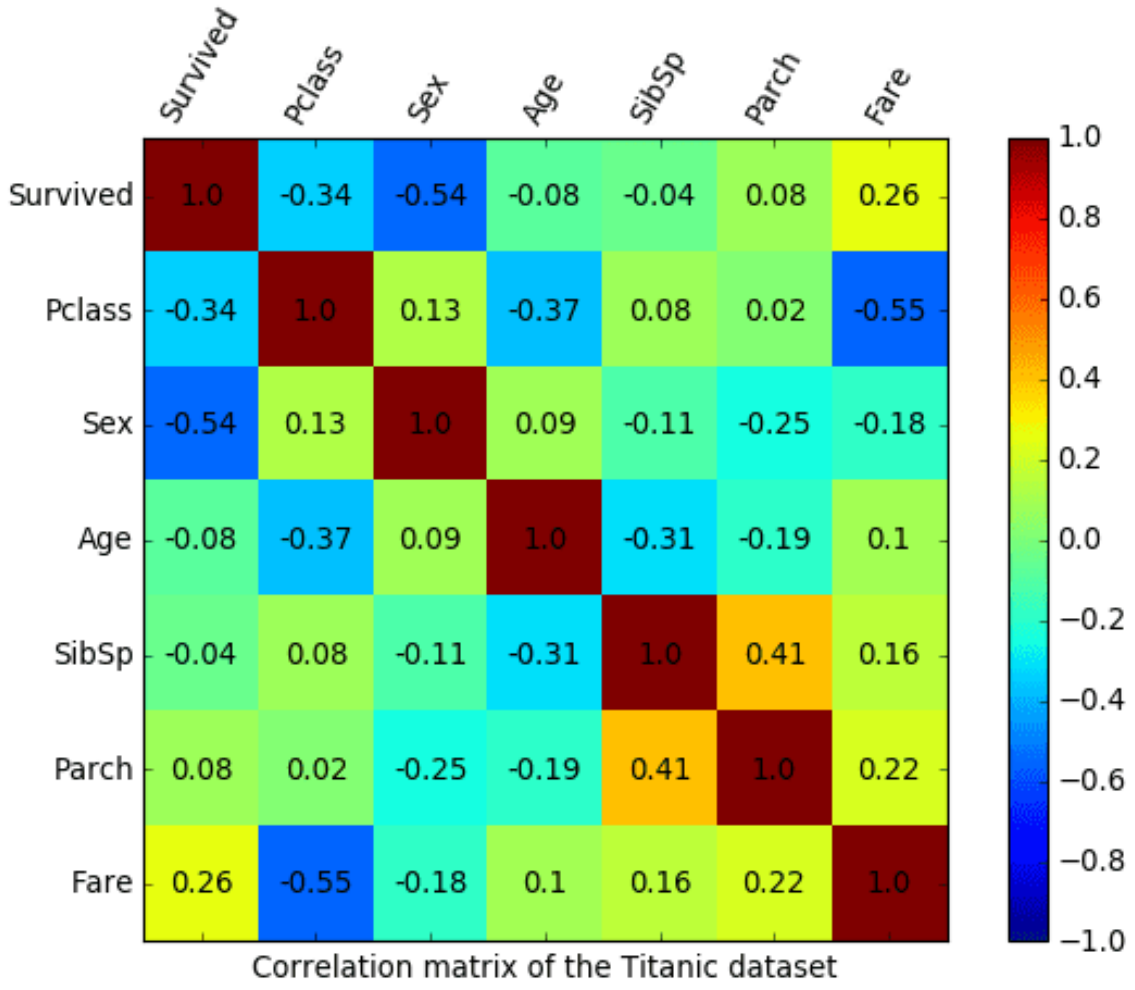
A dummy variable is a **numeric variable** that represents **categorical data**, such as gender, race, political affiliation etc... It only takes the value 0 or 1 to indicate the absence or presence of said effect that may be expected to shift the outcome.



2 - Data exploration

Correlation matrix

- Also known as a “heatmap”
- Shows **correlation** between all possible variable pairs
- Displays in a matrix form
- “How well do two variables explain each other?”
- Central axis is always “1”
- Both sides are mirrored images of each other
- Close to 1 or -1 → high correlation
- Close to 0 → low correlation



Interaction effect

- Occurs when the effect of one variable depends on the value of another.
- Captures **combined influence** of variables, beyond their individual effects.
- Often represented by multiplying two variables (e.g., ‘Years of education’ x ‘Seniority’).
- Useful for modeling more complex relationships in data.
- Helps reveal if variables together have a stronger or different impact than separately.

3 – Dimensionality reduction

Dimensionality Reduction

- Sometimes our data is too big to store/to explore manually
- Ideally, we want to keep all data points
- Solution: **reduce** the amount of features
  - = dimensionality reduction
- Ideal scenario: no information loss
- Realistic scenario: minimal information loss



Supervised or unsupervised?

Both!

Feature selection & feature engineering

- Feature selecting = **selecting** the most important (or least redundant) features of a dataset
- Feature engineering = **creating** new features based on multiple original ones
- Reduces size and variety of dataset
- Improves processing time when creating ML models
- Finding a balance between dataset size and model accuracy

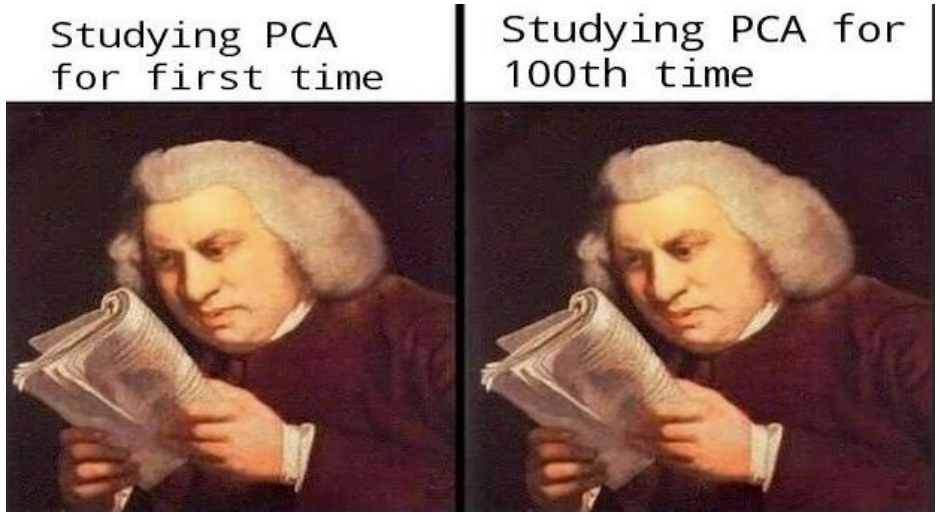


Lasso regression

- “Least Absolute Shrinkage and Selection Operator”
- Start with linear regression: minimize prediction errors
- Add a **penalty**: shrink large coefficients down
- Penalty strength controlled by parameter ‘**alpha**’ (α)
- Lasso forces irrelevant features' coefficients to zero
- Important features stay; unimportant ones are removed
- Final model is simpler and avoids overfitting.
- Finding the best alpha: draw a plot or even better, use **Grid Search Cross Validation**!

PCA

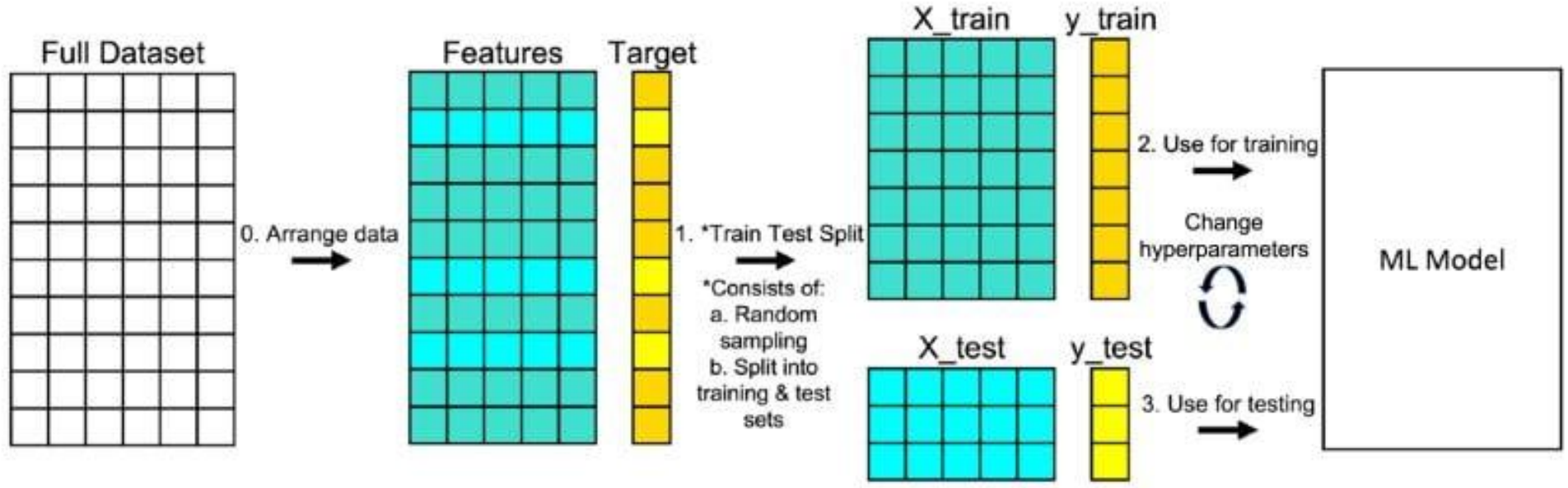
- ‘Principal Component Analysis’: reduce the variables to a few ‘principal components’
- Only one of the many dimensionality reduction techniques available
- Bundle variables with **high correlations** together
- Create a new variable to replace the two previous ones
- Try to keep as much explained variance in the dataset as possible
- Repeat until satisfied
- Figure out how many components to keep by plotting the **eigenvalues**!



4 – Model preparation

Train – Test split

By creating a train and test split we can train our model on certain data and test it on unseen data. This helps **combat overfitting** as we get a more accurate reflection of how our model performs when presented with new data.

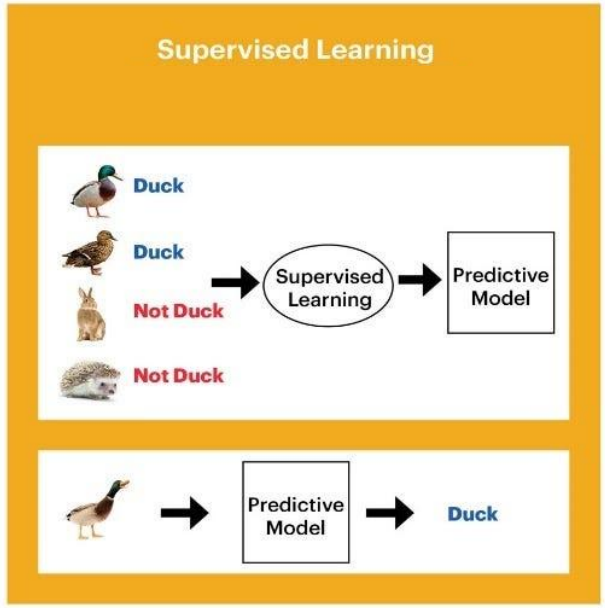


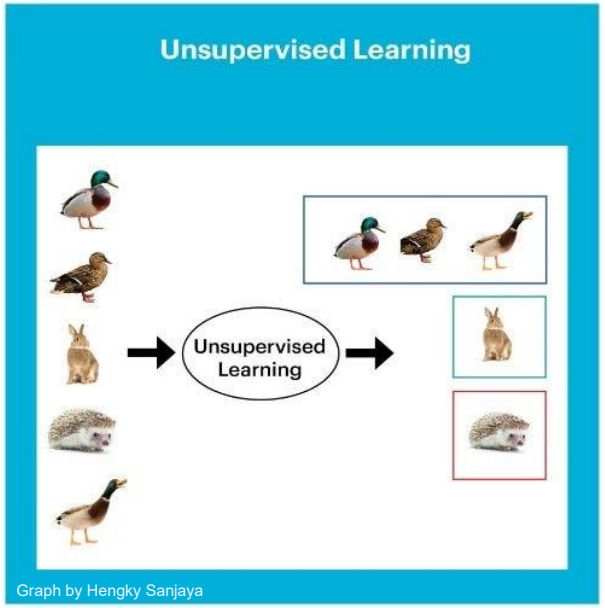
Source: Michael Galarnyk (builtin.com 2022)

5 – Model training

Types of machine learning

- Supervised
  - A ML task where the program is provided with training data that is labeled.
  - Given the inputs, the system knows what the expected output label is.
- Unsupervised
  - A ML task where the program is provided with training data that is unlabeled.
  - Given the inputs, we do not know what the output should be.





Western Digital.

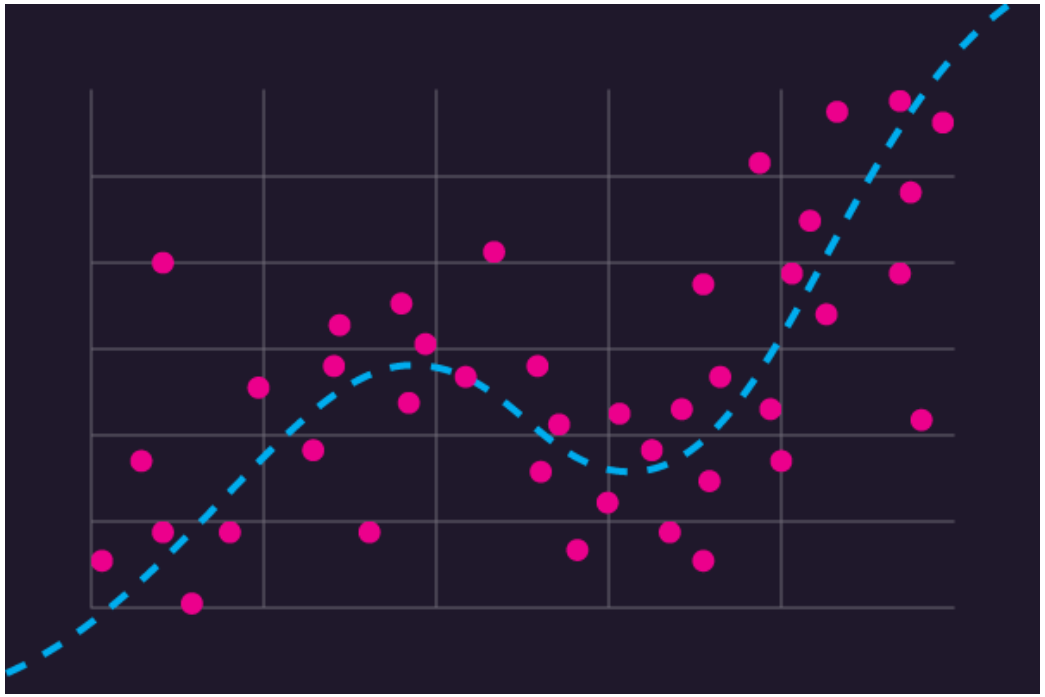
5.1 – Regression

Regression

- A method of machine learning wherein the variable we wish to predict (the dependent variable) is of a continuous nature.
- The value of a given entry is determined based on known cases.

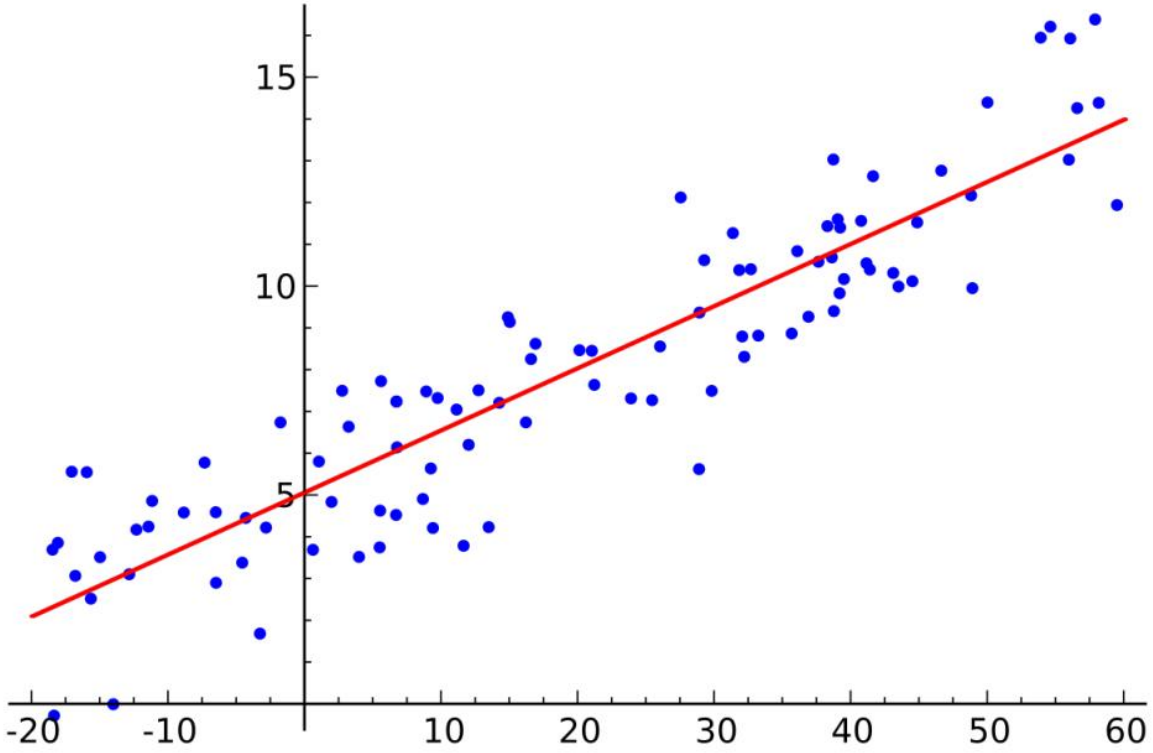
Supervised or unsupervised?

Supervised!



Linear regression

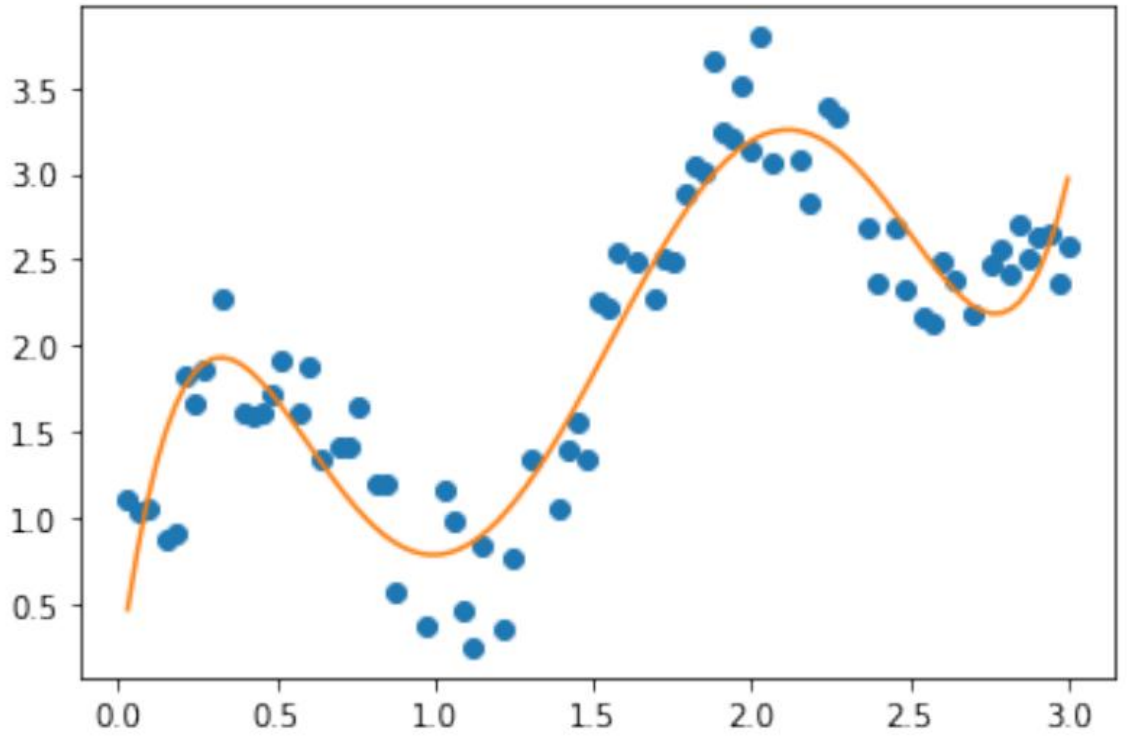
- The simplest form of regression
- A linear model → a straight line through the data
- The higher X, the higher (or lower) Y
- "line of best fit"
- Also works with multiple dimensions



$$y_i = \beta_0 + \beta_1 x + \beta_2 x + \beta_3 x + \beta_4 x + \ldots$$

Polynomial regression

- The ‘upgraded’ form of regression
- Linear regression is cool, but limited.
- Your model moves with the data
- “line of best fit”
- Also works with multiple dimensions



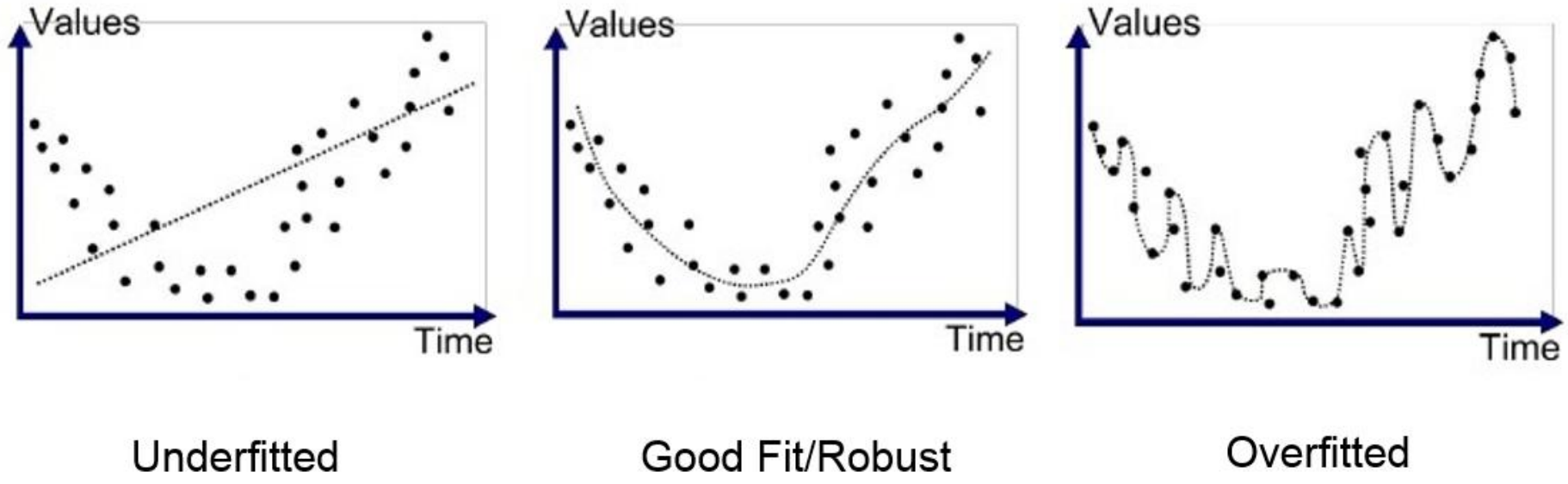
$$y_i = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 x^4 + \ldots$$

Evaluation metrics

- How do we know whether our model is any good?
- We have seen multiple metrics
- Look at the model itself:
  - R^2
- Compare your model to other models:
  - Mean Absolute Error (MAE)
  - Mean Squared Error (MSE) → preferred!

Overfitting

- The best way to combat overfitting in regression is by looking at the **MSE** on training versus testing data.
- Go for a high MSE and a low polynomial degree



Source: Anup Bhande, What is underfitting and overfitting in machine learning and how to deal with it. (Medium 2018)

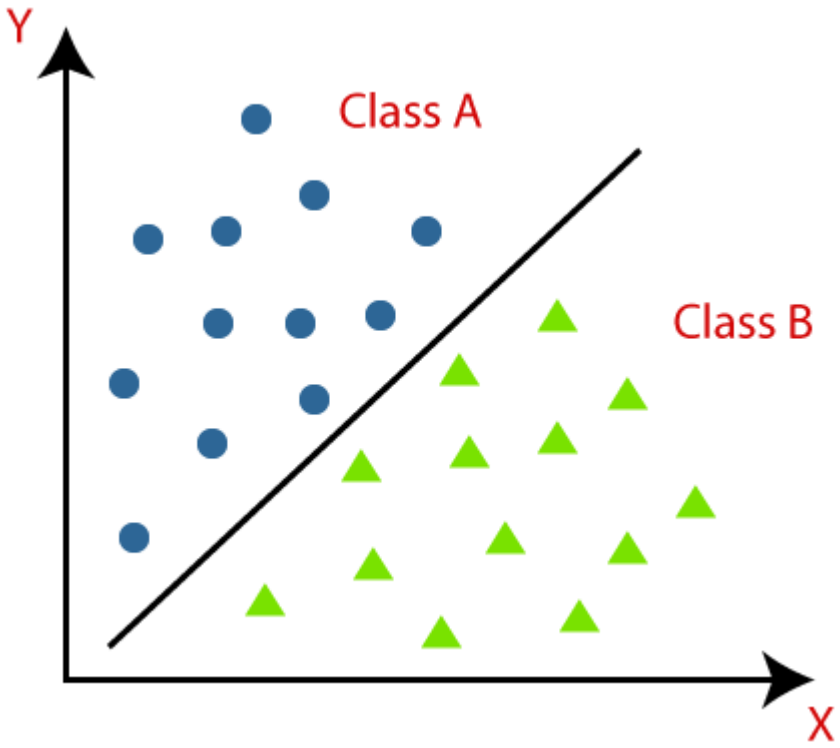
5.2 – Classification

Classification

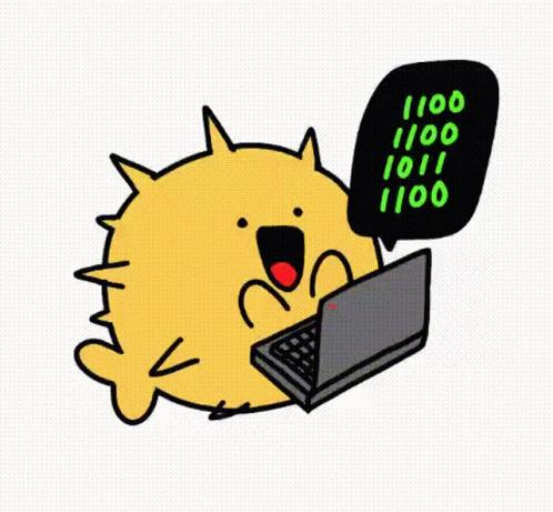
- A method of machine learning wherein the variable we wish to predict is of a categorical nature.
- The label of a given entry is determined based on known labelled cases.

Supervised or unsupervised?

Supervised!



Classification types

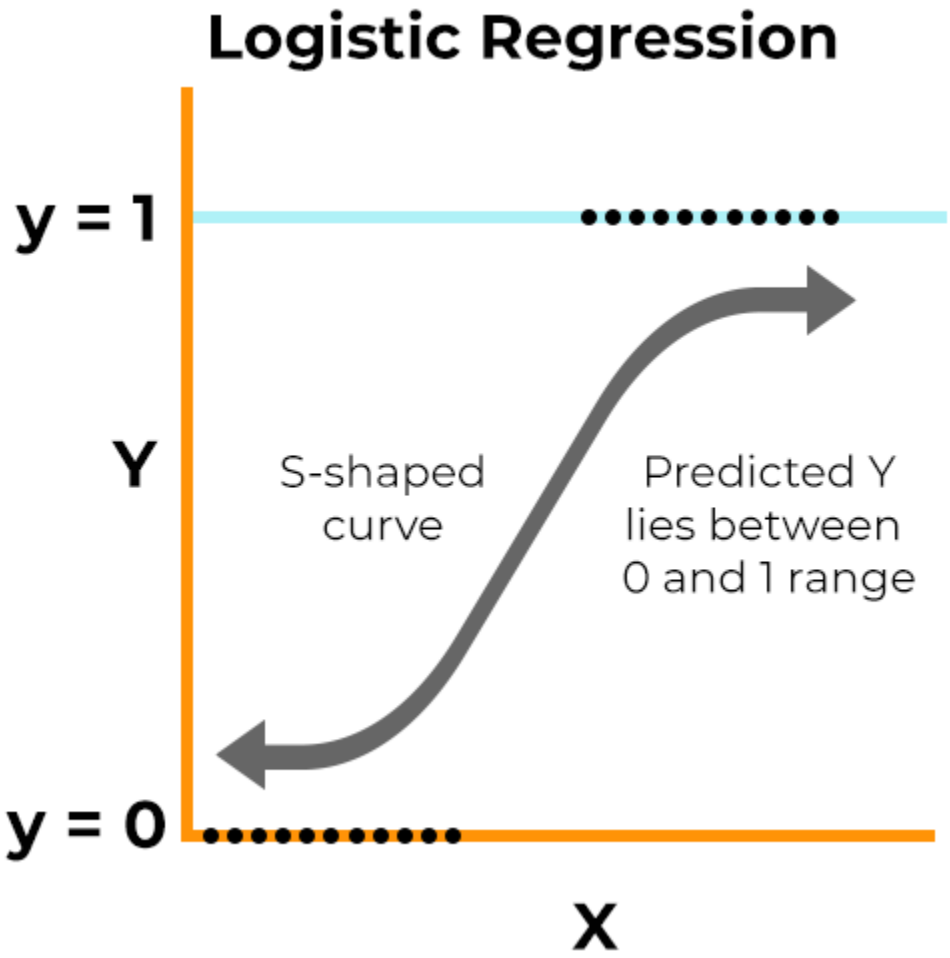


- Binary classification
  - Always 0 or 1
  - Predicts if a certain factor is True or False
    - E.g. Spam, Cancer, …
- Multiclass classification
  - Can be anything
  - Predicts the type of a certain factor
    - E.g. Wine quality rating, …



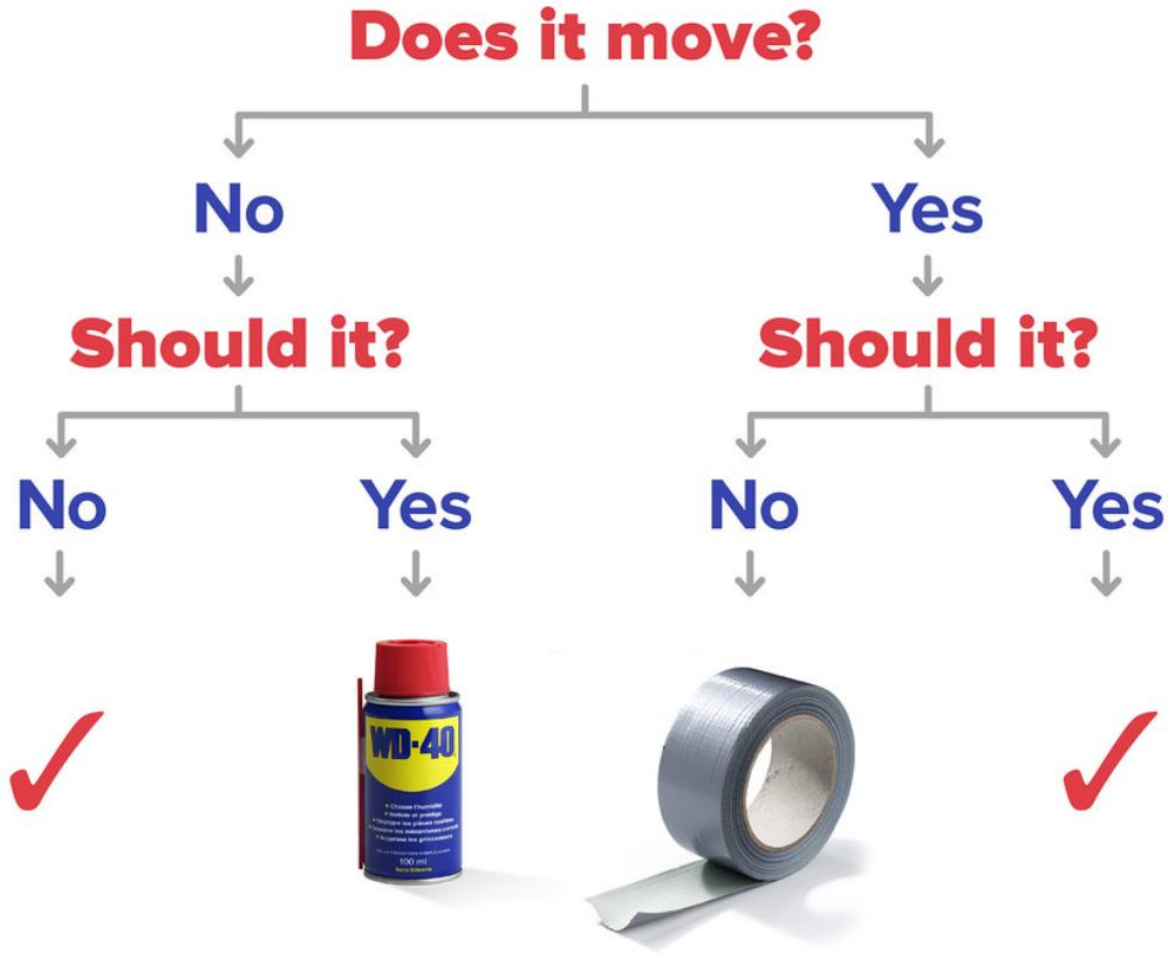
Logistic regression

- In its core a form of regression
- S-shaped curve
- Based on X, we predict Y
- Y always lies between 0 and 1
- This interesting property allows us to map each outcome → classification!
- Python also allows us to use logistic regression as multiclass classification



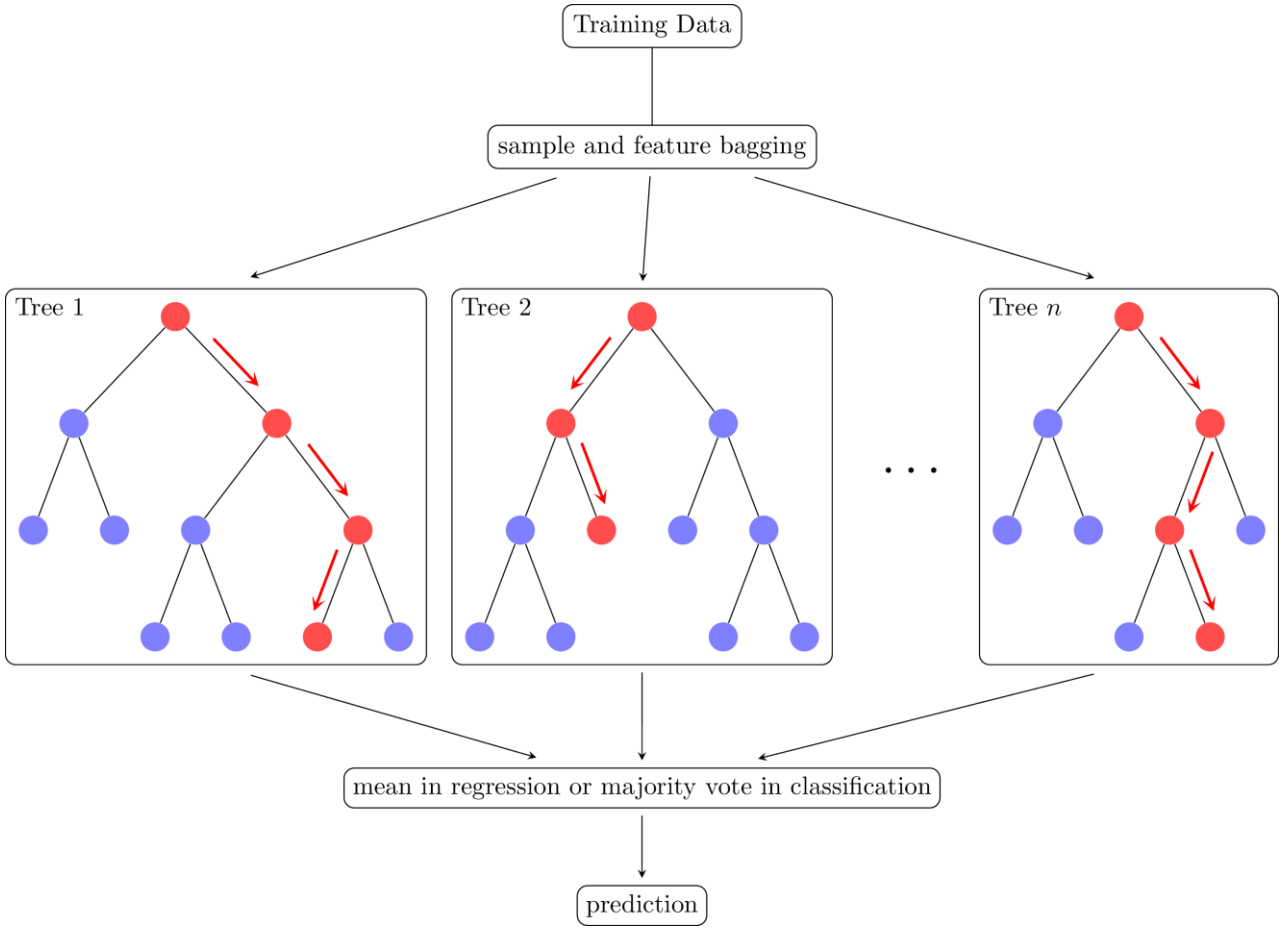
Decision trees

- Basically a flowchart
- Mimics human decision making
- Sever ‘splits’ where you ask a question
- Based on the answer you ask further questions
- Keep going till you end up with your answer!



Random forest

- Extended form of a decision tree
- Chain of decision trees
- Each tree:
  - Gets a subset of data
  - Makes a prediction
- Majority vote
- Can also be used for regression!



knn

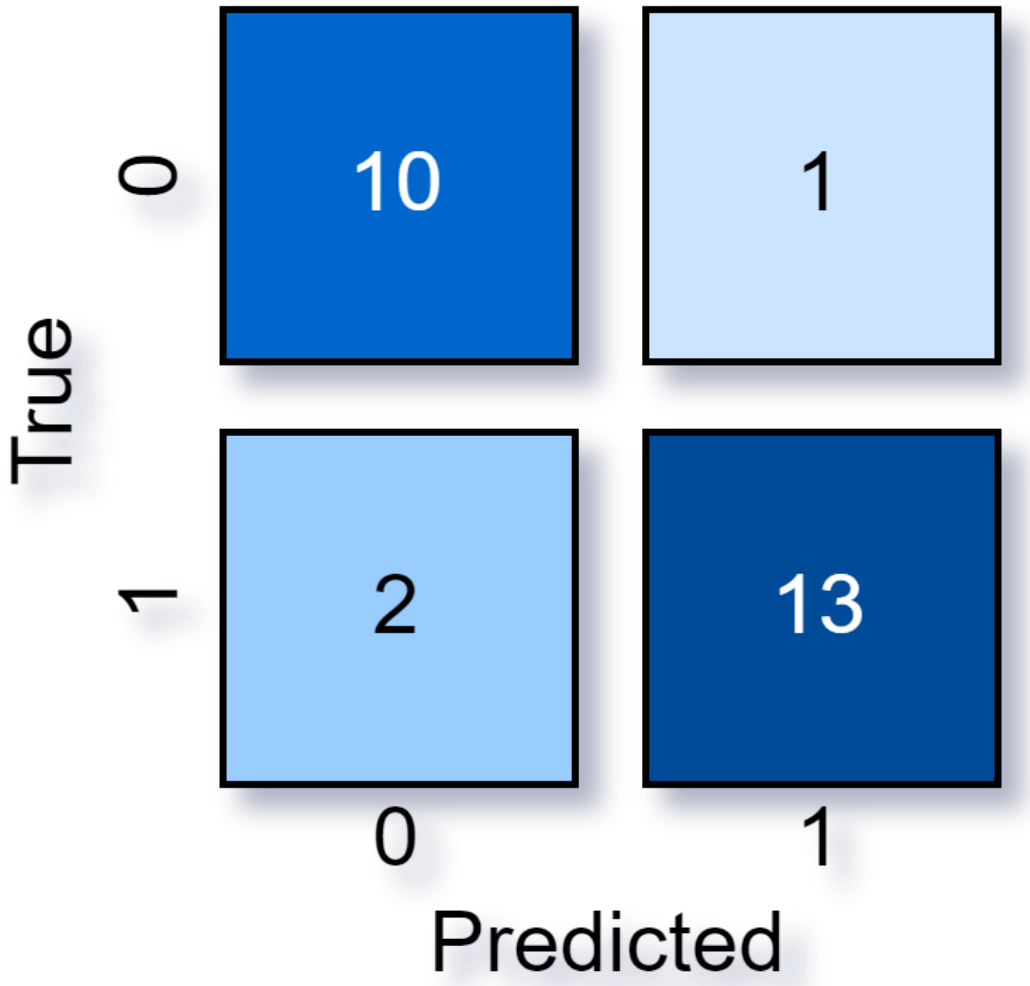
- ‘k nearest neighbours’
- Finds similar datapoints
- Each datapoint ‘votes’ on a class
- Majority vote: class with most votes wins!
- ‘k’ represents the amount of closeby datapoints to consider

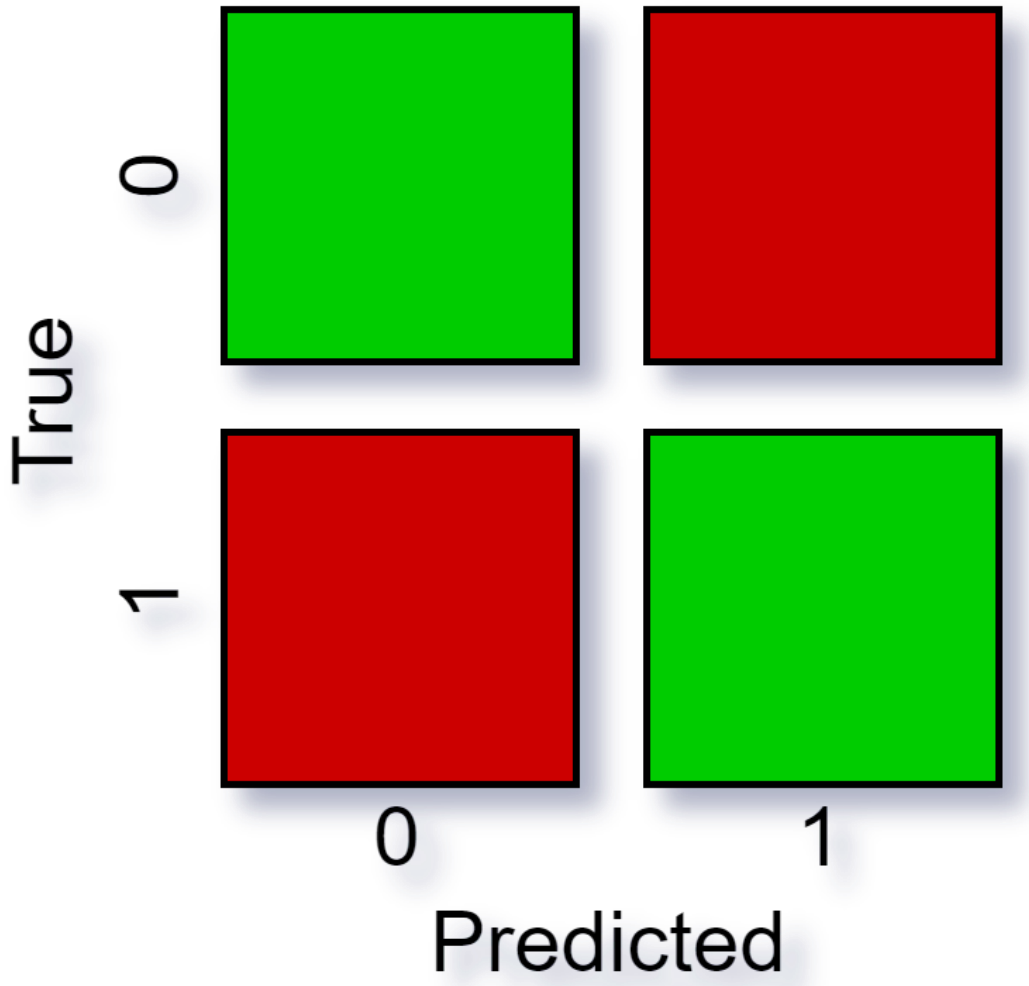


Evaluation terminology

- **True Positives (TP)**: The amount of labels where the outcome is 1 and the predicted label is 1. In other words, the amount of variables where we *correctly* predicted the postive outcome.
- **True Negatives (TN)**: The amount of labels where the outcome is 0 and the predicted label is 0. In other words, the amount of variables where we *correctly* predicted the negative outcome.
- **False Positives (FP)**: The amount of labels where the outcome is 0 and the predicted label is 1. In other words, the amount of variables where we *incorrectly* predicted the postive outcome.
- **False Negatives (FN)**: The amount of labels where the outcome is 1 and the predicted label is 0. In other words, the amount of variables where we *incorrectly* predicted the negative outcome.

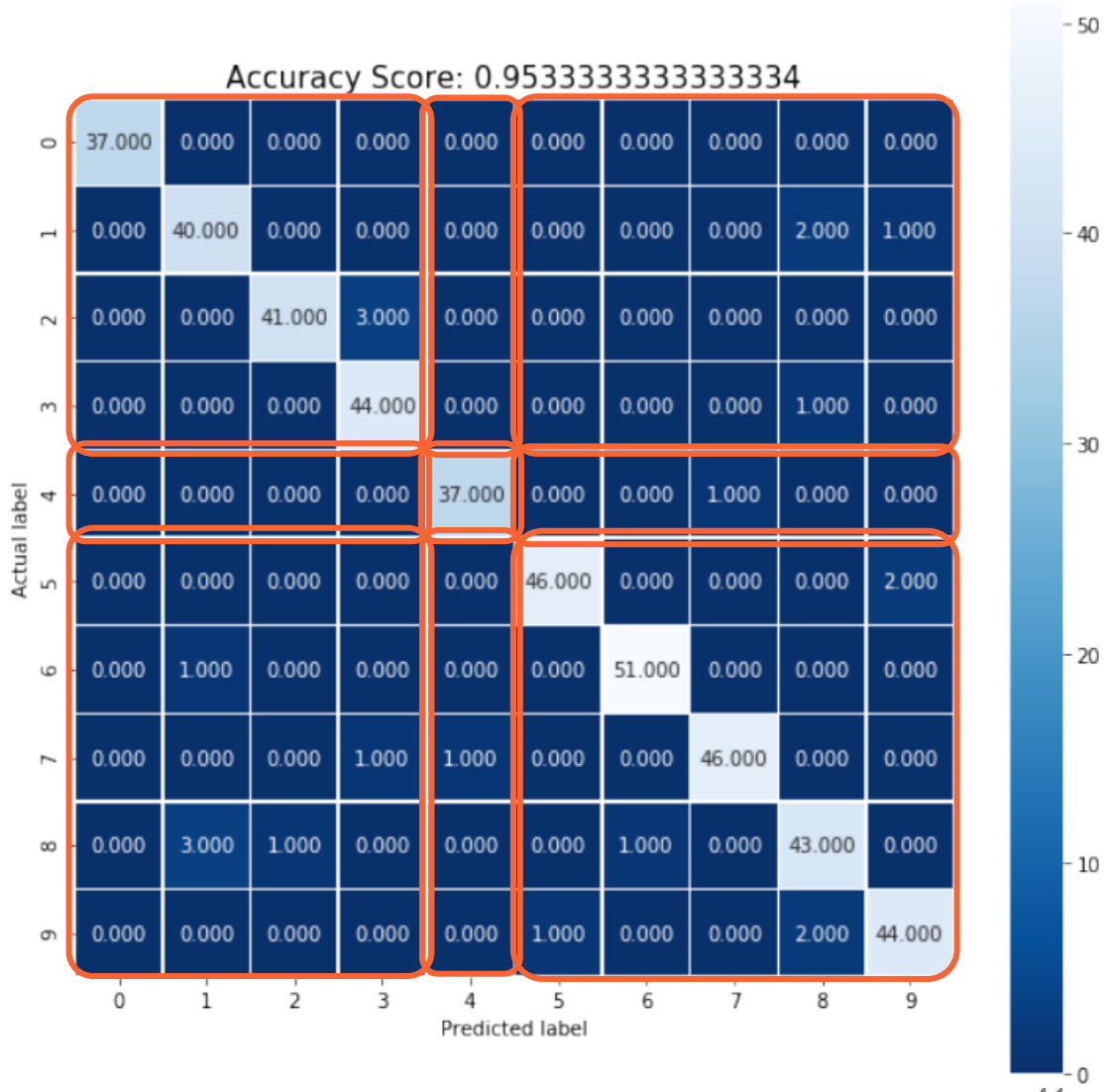
Confusion matrix





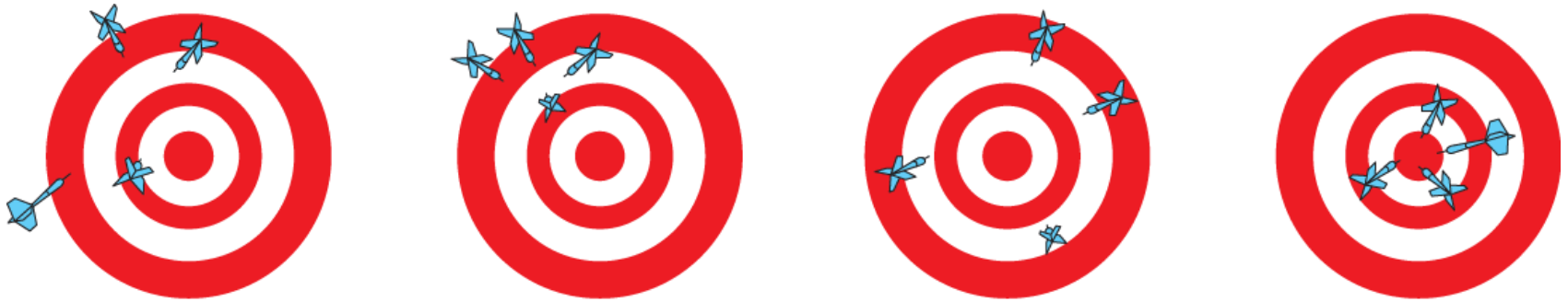
Confusion matrix

- The same goes for multiclass classification
- Calculate TP/TN/FP/FN per class
- You create an overal idea of your model by calculating an average
- Python can do both for us :D



Evaluation metrics

- Accuracy - “How correct are my predictions?”
- Precision - “How consistent are my predictions?”
- Recal - “Of all positive values, how many did my model recognize?”
- Specificity - “Of all negative values, how many did my model recognize?”
- F1-score: Gives a more balanced overview than accuracy



Combat overfitting

- Learning trends > recognizing values
- Decision trees:
  - The **fewer questions** a model asks, the **more** it needs to **learn** from those questions
  - Keep amount of nodes low
- Knn:
  - It tends to overfit on low values of k → doesn't learn patterns, just remembers some data
  - On high values it tends to underfit → doesn't learn patterns, just does the same for each datapoint
  - The best value is somewhere in between, so we rely on other methods (see 6 –Evaluation methods)

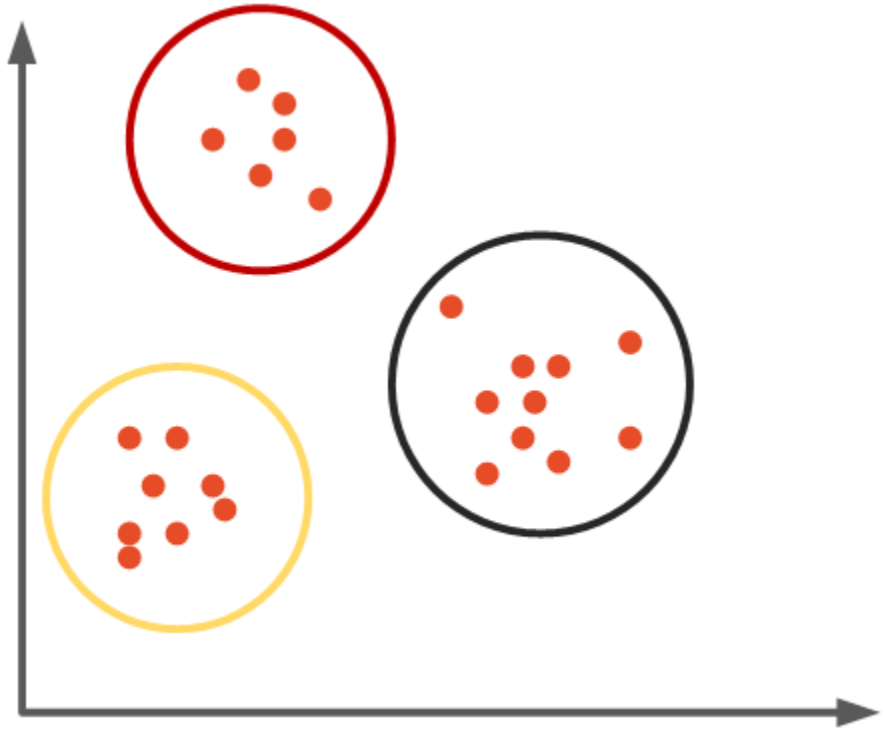
5.3 – Clustering

Clustering

- A method of machine learning in which we try to detect clusters of observations in our dataset.
- A given entry is assigned to a group base on the entire dataset.

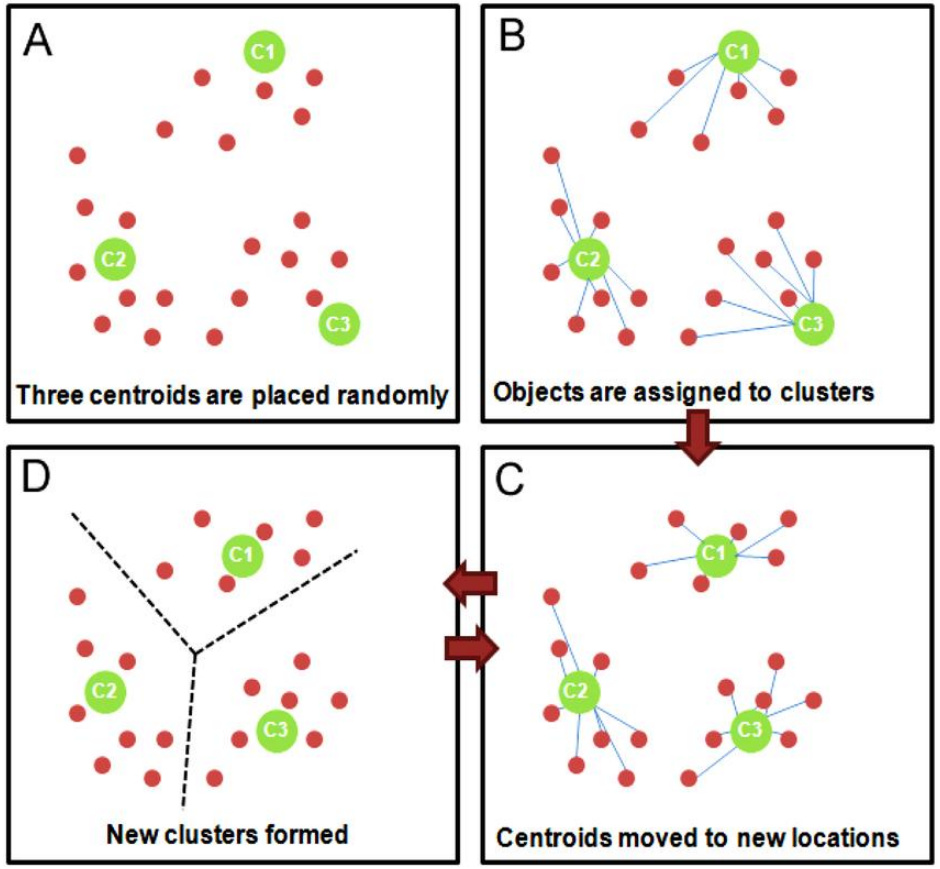
Supervised or unsupervised?

Unsupervised!



K means

- Basic clustering algorithm
- Start with a parameter ‘k’ = amount of clusters
- Assign a random starting point to each cluster (= ‘centroid’)
- Assign each datapoint to nearest centroid
- Move cluster to middle of each assigned datapoint
- Repeat until converged

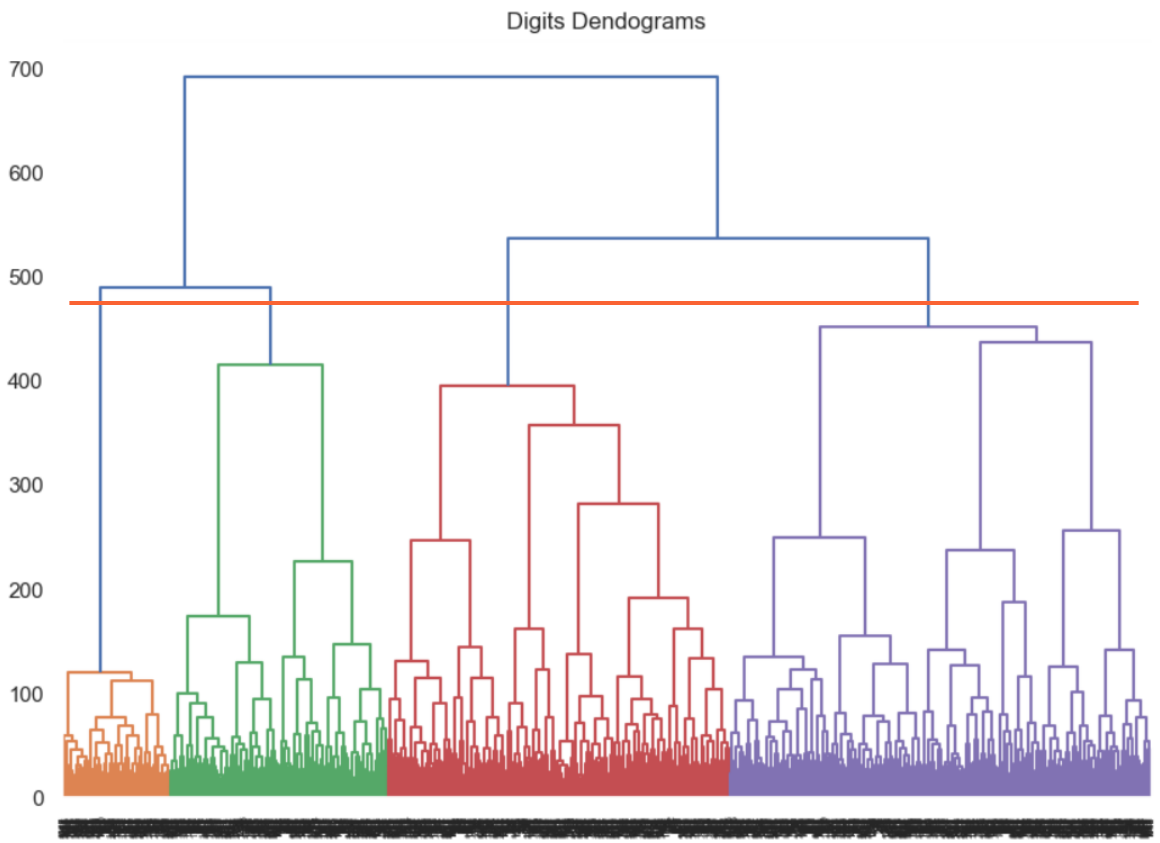


Hierarchical Clustering

- More advanced clustering method
- Flexible approach to choosing the amount of clusters
- Bottom-up mechanic
  - each individual point starts off as a unique cluster, at the end there is only 1 cluster left.
- Clusters are formed by merging the clusters which are closest together.

Dendrogram

- Visual representation of **all** clusters
  - From max to 1
- X-axis show the datapoints
- Y-axis shows the distance between clusters (Linkage Criterion)
- Lines show the grouping of clusters
- Define colours for each cluster (optional but nice :D)



Evaluation metrics

- Since we have no labels, there is no way to evaluate our model
- The only way to confirm is by plotting your data
- If you limit you data to two columns, you can create a graph

- Secret technique: we can ‘mold’ our clustering model to produce results



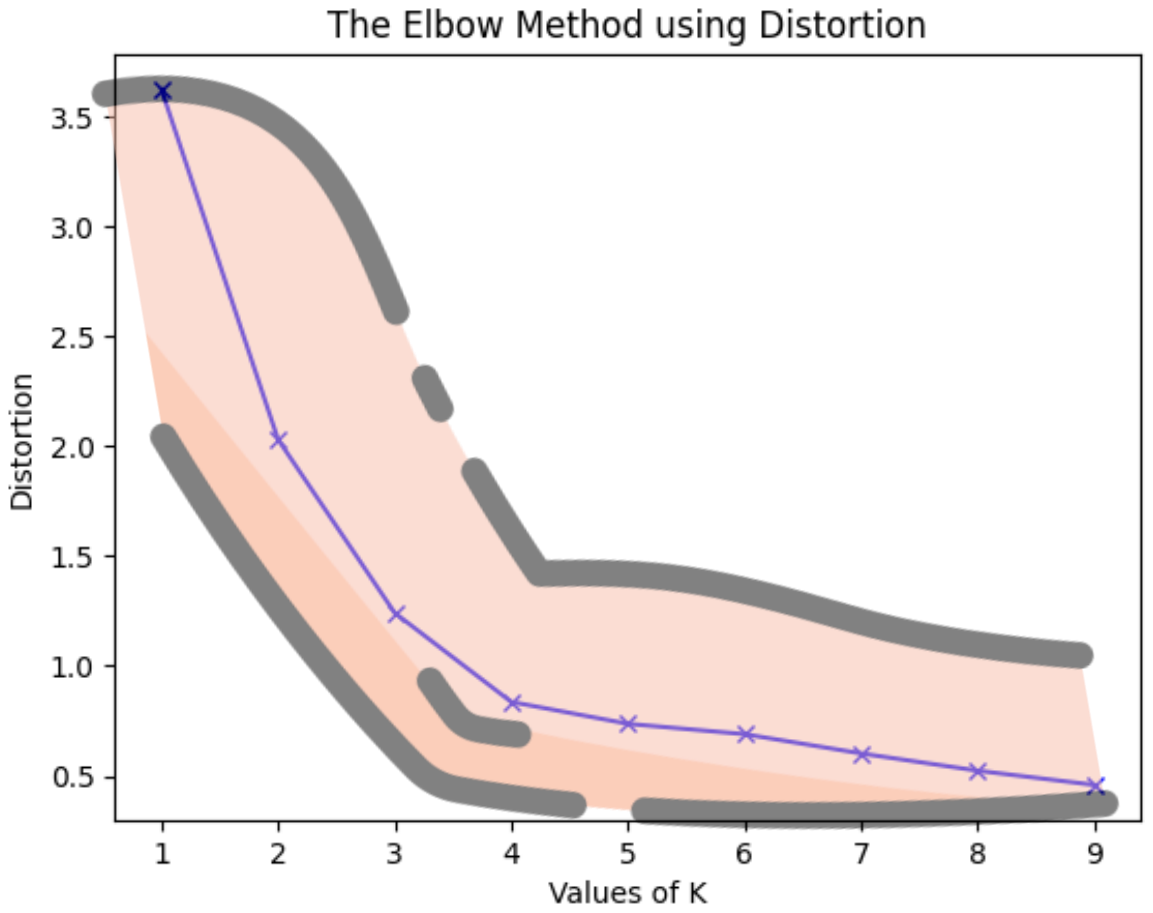
Clustering as a predictive model

- Strip your dataset of labels
- Perform clustering
- Assign each datapoint to a cluster
- Map each cluster to a label
  - Look at which class each cluster is **most likely** to represent
  - =/= 100% reliable!
- Compare ‘predicted’ label with stripped label
  - Accuracy, precision, correlation matrix, …
  - Clustering → no need for a train test split!
- Done!



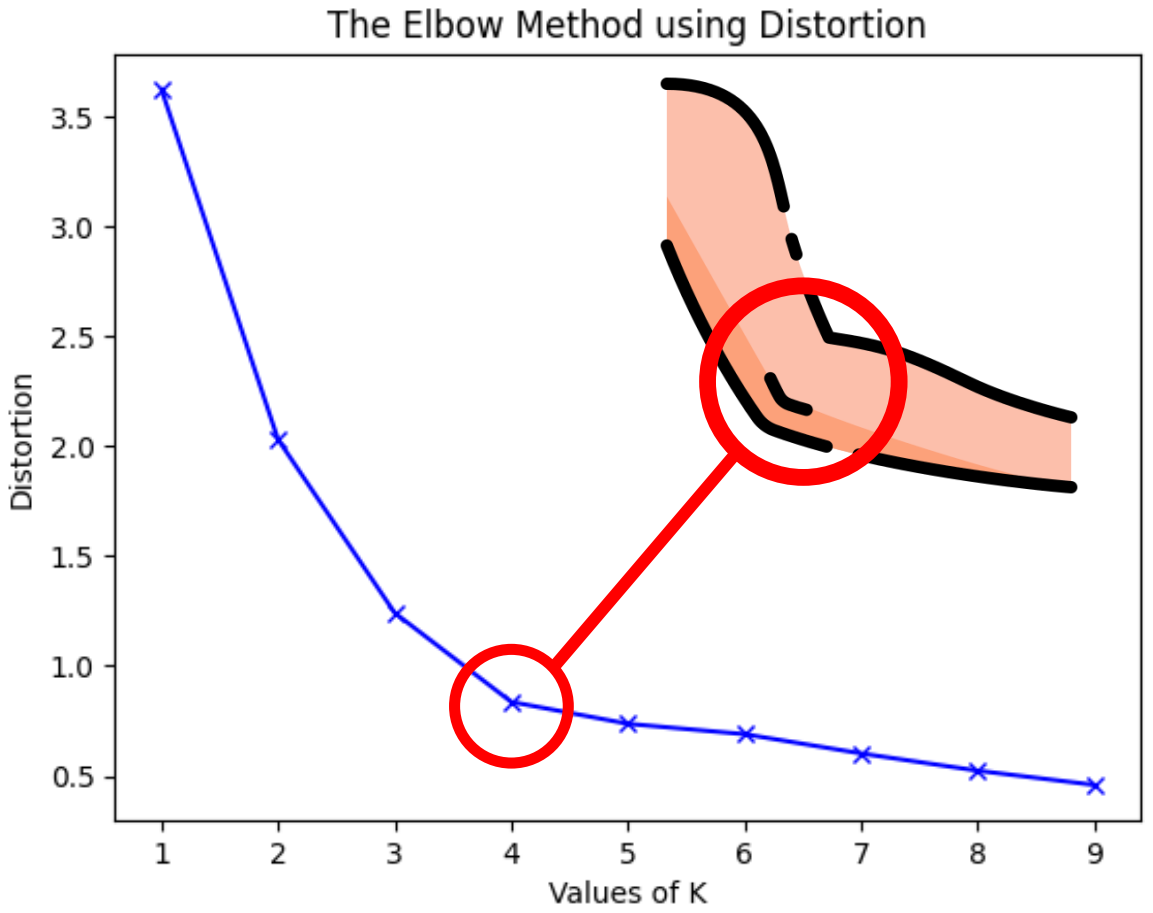
Elbow method

- A way to combat overfitting in clustering
- Train a bunch of models
- Calculate '**distortion**' per model
- = average squared distance between each datapoint and its nearest centroid
- → use **inertia** for hierarchical clustering
- Find the right **balance** between low **distortion** and low value of **k**
- Pick the value in the middel of the 'elbow'



Elbow method

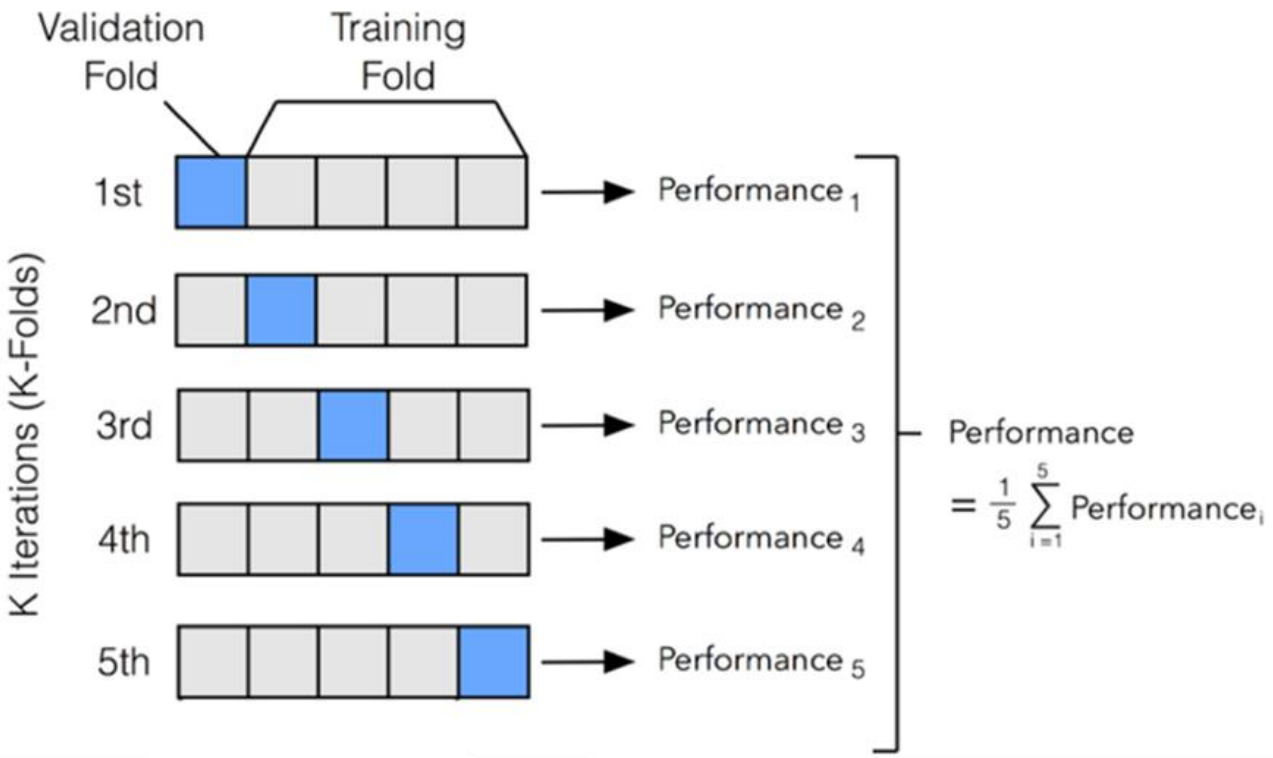
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6 – Evaluation methods

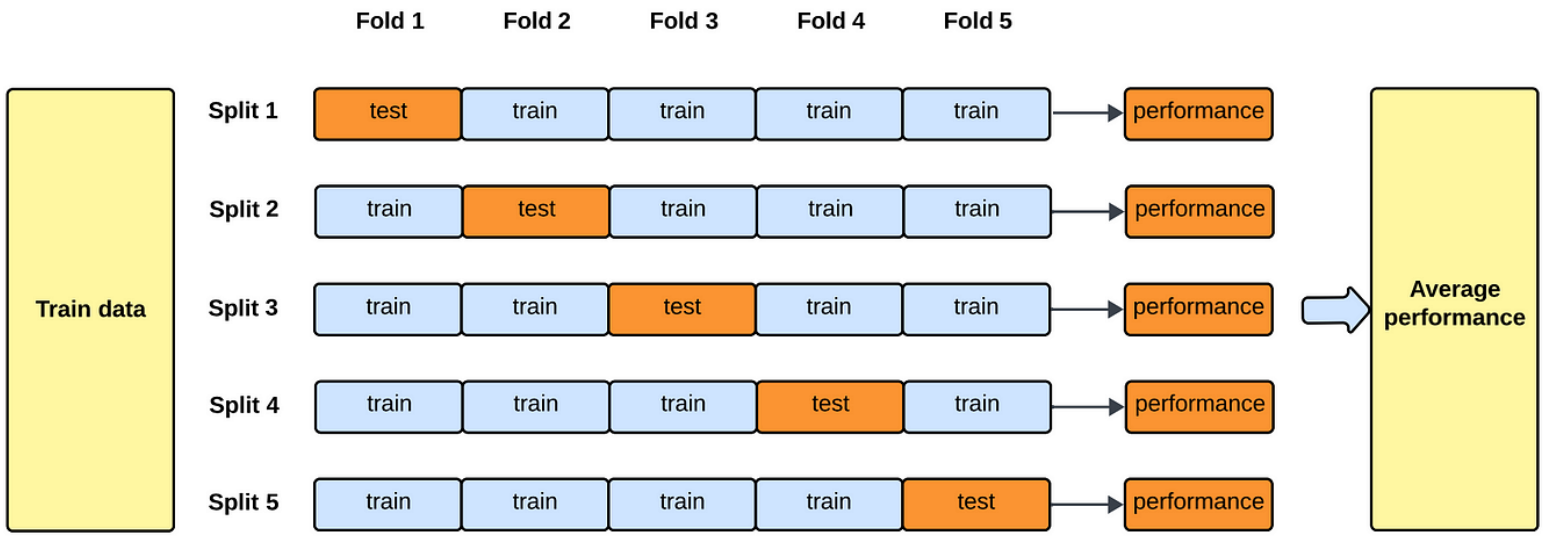
Cross validation

- Used to compare different models
- Create ‘folds’ in the data
- For each fold, train a model with that fold as test set
- Take the average outcome: classification → majority vote!
- Special cases:
  - ‘Holdout’
  - ‘leave-one-out’
  - ‘Stratified’



K-fold

- ‘k’ represents the amount of folds
- Most common model: 10-fold
- The value of ‘k’ cannot be higher than the amount of datapoints!



Source:Haden Pelletier, How-To: Cross Validation with Time Series Data (Medium 2023)

Take a closer look at what we've been working towards

Exam preparation

Exam DSF

- Oral exam
- About 15-30 minutes
- Focus lies on your **understanding** of the material, not your **application** of it (you'd just use AI tools for that in a real job anyways)
- We start from a few **key questions**, and build a **conversation** from there



Content



- What does this mean?
  - Theory
  - Inner workings of models
  - Pitfalls to look out for and how to solve them
  - Formulas
  - Specific applications
  - …
  - Anything we've covered in the slides, exercises or orally during classes
- **No strict coding questions**
  - You will NOT be asked to write any code
  - You MAY be asked to analyse and interpret already written code and graphs

General advice



- Do not stress! Here's why:
  - If you know the material, it will be a breeze
  - Oral exams are **easier** than written exams as you get a chance to correct yourself
  - I am here to test what you **learned** and where your **limits** are, not to make you fail
- Keep it simple, but not vague:
  - Ensure you **fully explain** everything I ask
  - But **don't get lost** in **irrelevant topics** where you might make a mistake!

- If you are able to explain the topics you your little brother/sister/… then you know it well

Practical flow

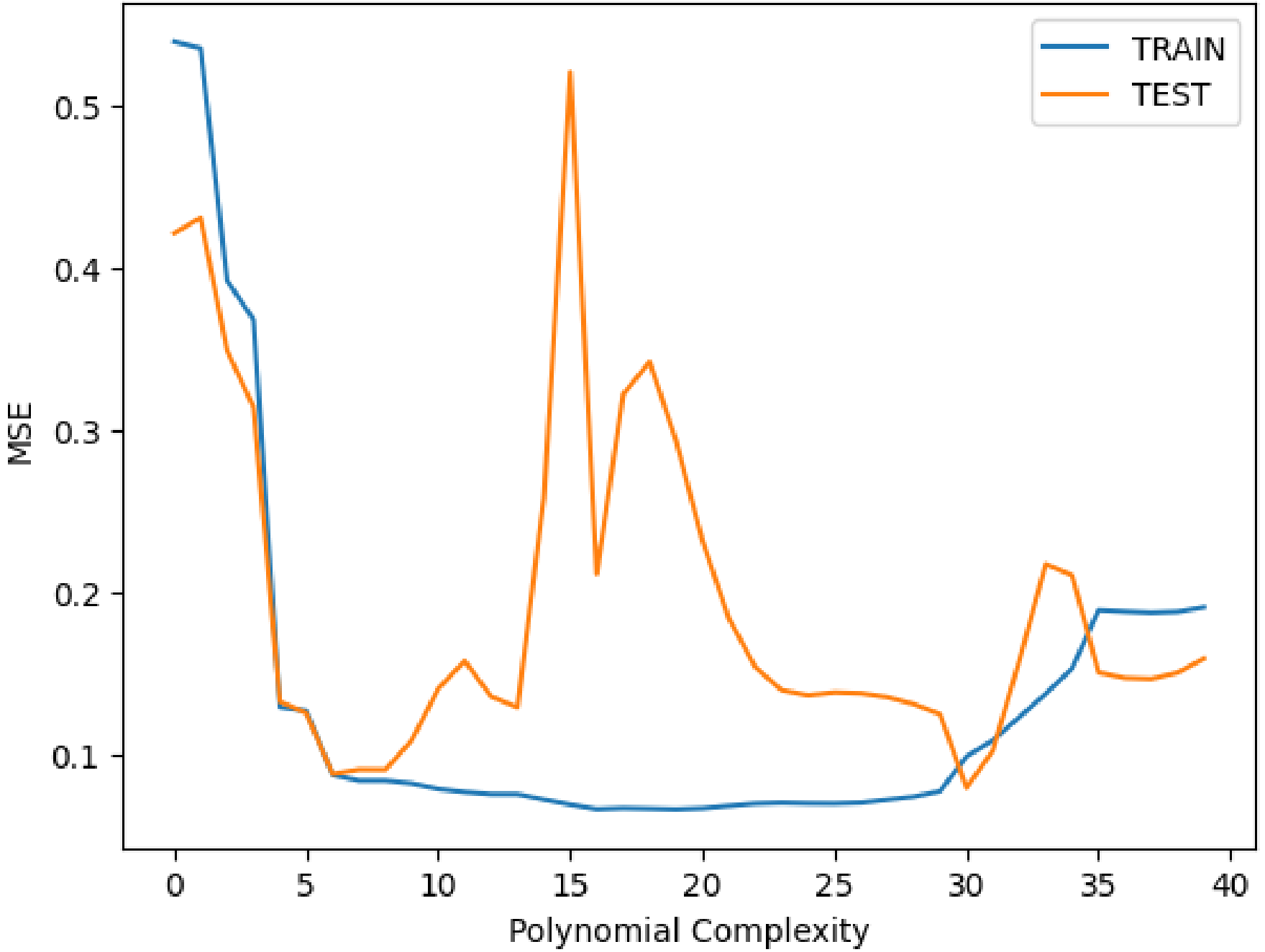
- Classroom: **C1.25** → Campus De Vest
- Be on time:
  - Find your timeslot at examen.thomasmore.be
  - Arrive at least **10 minutes** before your slot!
- Travel light:
  - You do **not** need to bring **anything**!
  - No laptop (remember: no coding questions)
  - No pen/paper (I will provide this if needed)



- Dress code:
  - No sweatpants/beach clothing/…
  - No need to go overly formal
  - Dress how you would for a normal schoolday!
  - Or go nuts and dress to make me laugh and ease the tension of the exam ;)

Sample questions

- Can you explain what logistic regression is and when or how it is used, including its limitations?
- What is a confusion matrix, and how does it work in classification problems?
- How can the KNN model overfit or underfit, and what factors contribute to these issues?
- How do I determine the best polynomial degree here?
- …


TRAIN
TEST
MSE
0.5
0.4
0.3
0.2
0.1
0
5
10
15
20
25
30
35
40
Polynomial Complexity